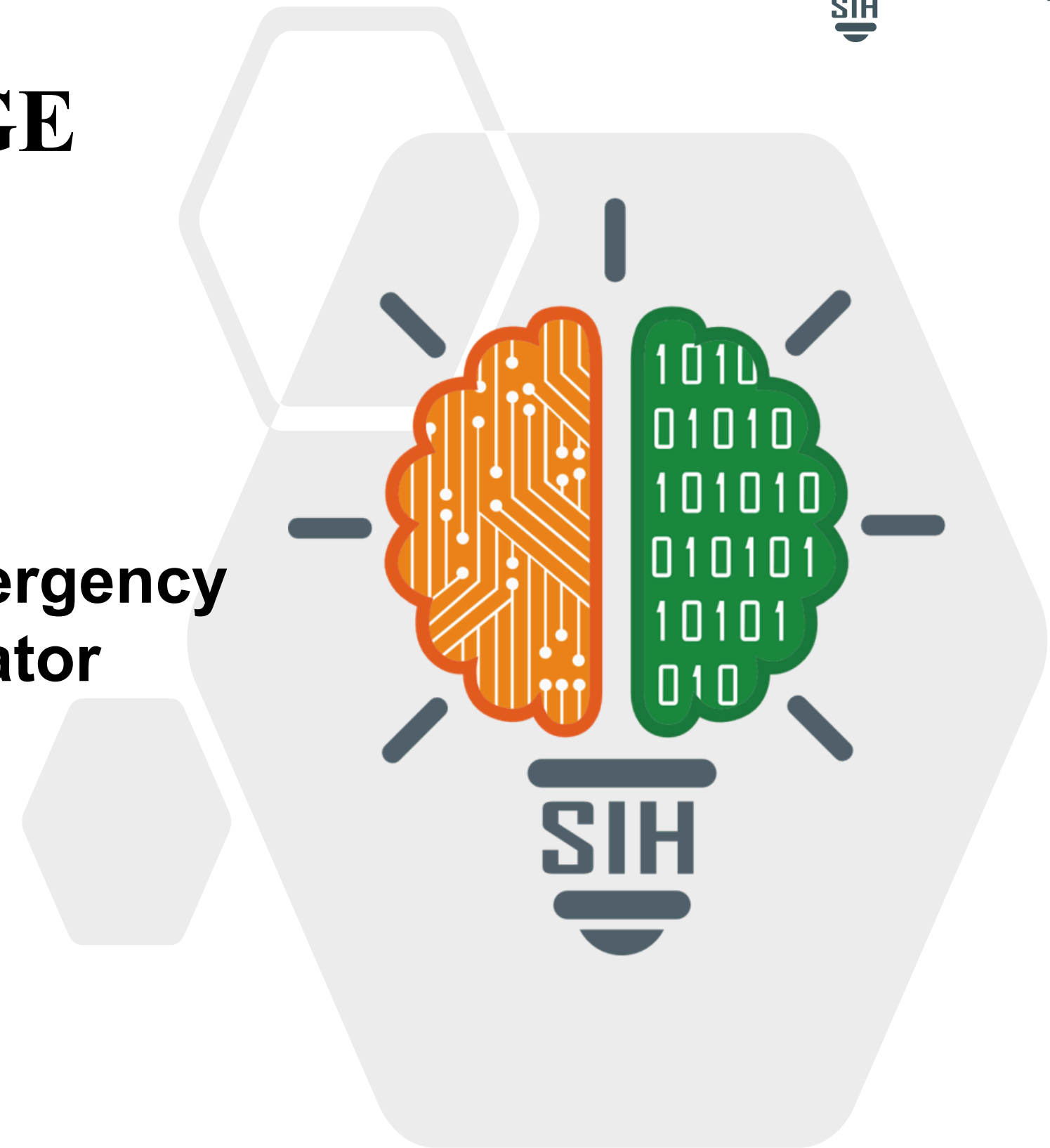


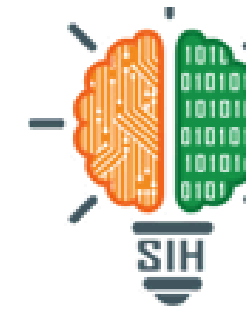
TITLE PAGE

- **Problem Statement ID –**
- **Problem Statement Title- Blackout Emergency Coordinator**
- **Theme- Disaster Management**
- **PS Category- Software**
- **Team ID-**
- **Team Name- GeoGuardians**





DisasterSafe



PROPOSED SOLUTION

- **Unified emergency response platform** connecting authorities, rescue teams, hospitals, volunteers and victims
- **Multi channel SOS** through Web, SMS and LoRa
- **Real time tracking** of victims, teams and resources using GPS
- **Offline communication** using ESP32 and LoRa when networks fail
- **AI assisted response** for alerts, triage, guidance and decision support

How it addresses the problem

- **Connects multiple agencies** on a single platform
- **Enables faster SOS alerts** and victim location tracking
- **Maintains communication** during network failures
- **Improves resource management** for teams and responders
- **Supports faster decision making** through AI assistance



LEVEL 1 ONLINE

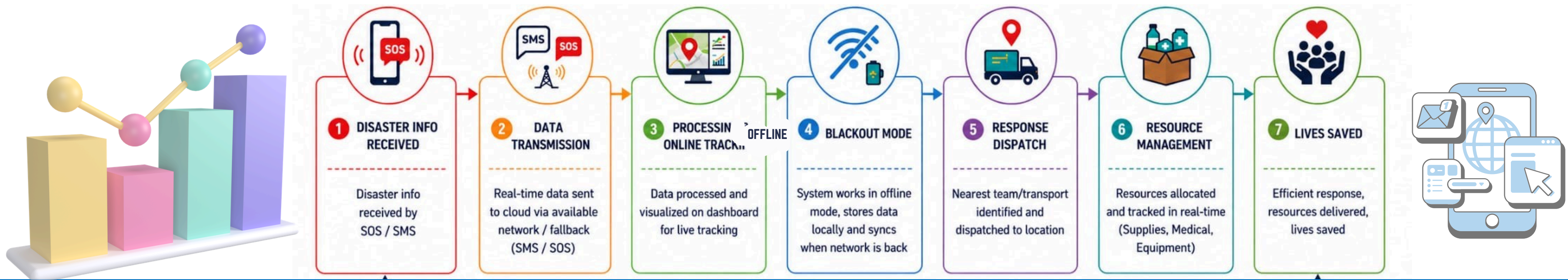
Languages: JavaScript / TypeScript
• Python
Frontend: React / React Native
Backend: Node.js / PHP • REST API
Data & Connectivity: MySQL • GPS/Maps • WebSockets • Live APIs
AI: LLM + RAG • Recommendation Engine

LEVEL 2 SEMI-ONLINE

Languages: JavaScript / TypeScript
Mobile: React Native
Backend: Node.js / PHP • REST API
Data: Local/Emergency Database
AI: Quantized LLM • Local RAG

LEVEL 3 OFFLINE

Languages: C / C++ • HTML / CSS / JavaScript
Hardware: Arduino Core / ESP-IDF
Local Server: ESPAsyncWebServer
Storage: LittleFS
Communication: RadioHead / LoRaWAN Mesh



Use Cases & Business Potential



Authorities

Manage teams, SOS, vehicles & resources.



NGOs

Coordinate volunteers and supplies.



Volunteers

Receive tasks, navigate & report updates.



Citizens

SOS, Smart offline assistant nearby help & emergency guidance

Business Potential

Gov/NGO Deployment: Disaster-response coordination platform.

Institutional Use: Campuses, industries & emergency response teams.

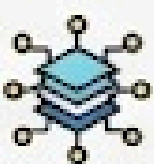
Scalable Model: Region-wise deployment & service integration.

Future Financials

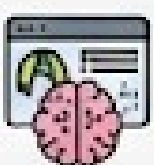
FUTURE FINANCIAL
PIE CHART
PLACEHOLDER



Feasibility



Technical: 3-level architecture supports online, semi-online & zero-connectivity operation.

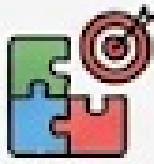


Operational: Role-based dashboards & AI assistant simplify rescue coordination and resource allocation.

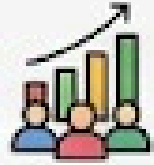


Economic: Uses open-source technologies, local storage and low-cost hardware.

Viability



Problem-Solution Fit: Enables emergency coordination even with limited/no connectivity.



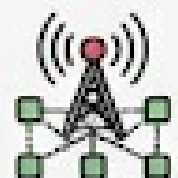
Market Need: Addresses communication, resource and rescue coordination gaps during disasters.



Scalability: Modular design supports more users, resources, regions and disaster types.

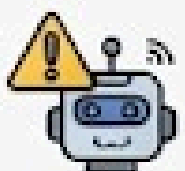
Challenges and Mitigation

Challenge: Network failure during disasters.



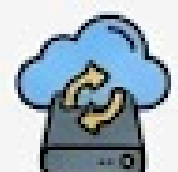
Mitigation: LoRa/mesh + local offline communication.

Challenge: AI accuracy in emergencies.



Mitigation: Quantized LLM + Local RAG with verified disaster guidelines.

Challenge: GPS/data synchronization.



Mitigation: Local caching + delayed synchronization when connectivity returns.

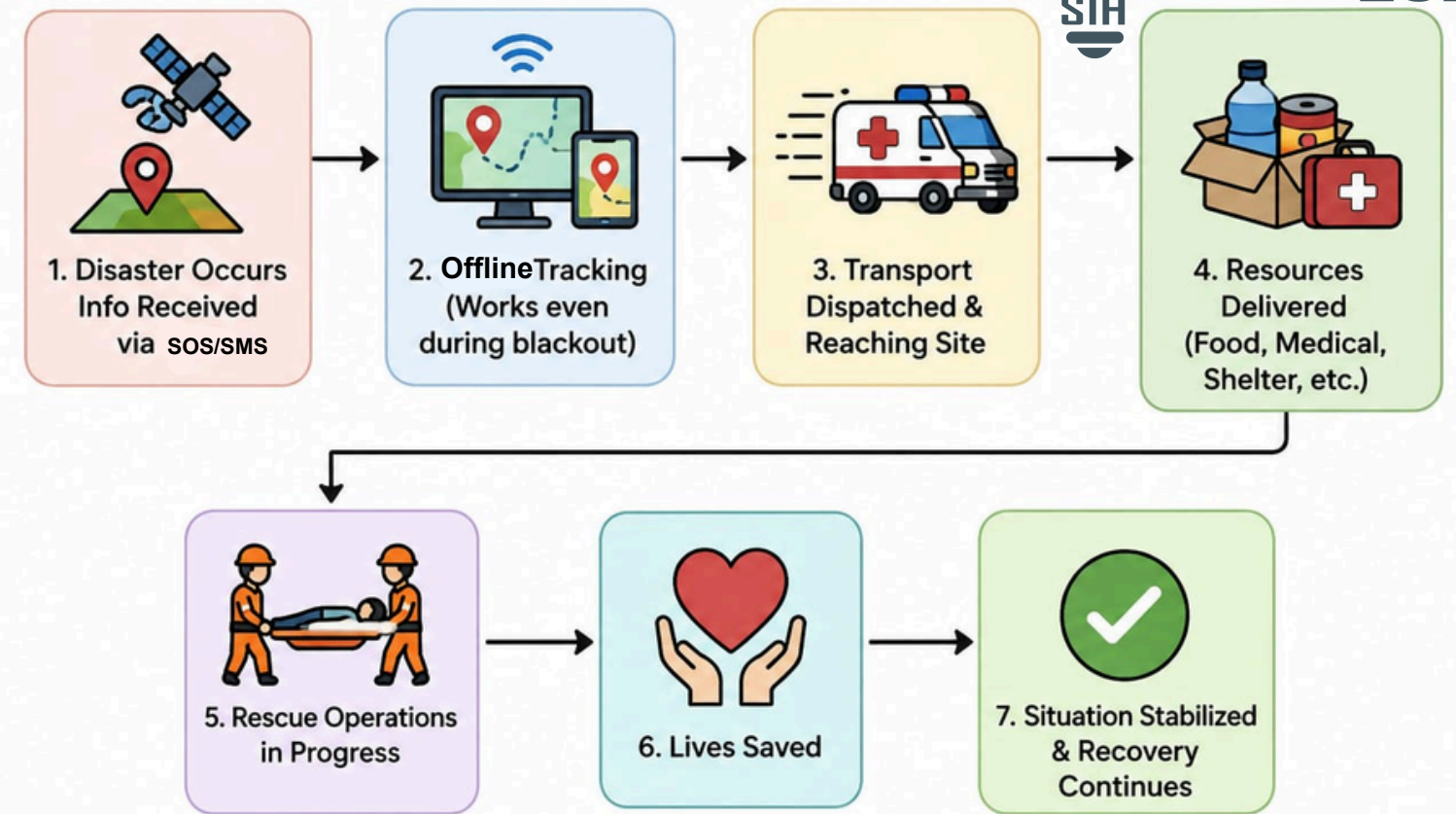
Potential Impact on Target Audience

- 🎯 **Victims / Citizens:** Faster SOS response, nearby help, shelters & emergency guidance.
- 🚒 **Authorities:** Better team, vehicle and resource allocation with live tracking.
- 🤝 **NGOs:** Efficient volunteer and resource coordination across affected areas.
- 👷 **Volunteers:** Clear task assignments, navigation and real-time field reporting.

IMPACT AND BENEFITS

Benefits of the Solution

- 🌐 **Social:** Faster rescue, better coordination & improved public safety.
- 💰 **Economic:** Reduces resource wastage, response costs & operational delays.
- 🌱 **Environmental:** Optimizes resource/vehicle deployment and reduces unnecessary movement.
- 🚑 **Emergency Response:** Faster SOS handling, live tracking & smart resource allocation.
- 📡 **Connectivity:** Continues functioning in online, semi-online & zero-connectivity zones.
- 🤖 **Technological:** AI-assisted decisions, offline intelligence & scalable architecture.



Platform	 Offline Functionality Works without internet	 SOS Transmission Sends help when network is weak/unavailable	 End-to-End Coverage From incident to resource deployment	 Guidance & Resources Actionable information for citizens
 DisasterSafe	 Full Offline Support Access guides, maps, and tools without internet	 Multi-mode SOS Works via Internet, SMS fallback & queue system	 Complete Workflow From incident reporting to resource deployment	 AI-Powered Guidance Contextual alerts, survival guides & resource info Offline AI assistance
 Existing Platforms	 Limited or No Offline Access Most features require active internet	 Internet Dependent Falls when network is unavailable	 Partial Coverage Focus on alerts only, limited follow-up flow	 Generic Information Standard alerts, limited personalized guidance

Crowdsourced Disaster Management — Citizen Data via Mobile Devices

<https://github.com/arpith20/Disaster-Management-Using-Crowdsourcing>

RescueRing — Disaster SOS, Mapping, Volunteers & Resources

<https://github.com/RescueRing>

Cicek & Kantarci (2023) — Crowdsensing in Disaster Management: Review & Challenges

<https://www.mdpi.com/1424-8220/23/3/1699>

Capability	112 ERSS	SACHET	IDRN	NDEM	DisasterSafe
Citizen SOS	✓	✗	✗	✗	✓
GPS	✓	✓	✓	✓	✓
Offline Survival Guide	✗	✓	✗	✗	✓
Emergency control system	✓	✓	✗	✓	✓
GIS map	✓	✓	✓	✓	✓
Resource inventory	✓	✗	✓	✓	✓
Resource assignment	✓	✗	✓	✓	✓
Multi-agency	✓	✓	✓	✓	✓
Communication fallback abstraction	✓	✓ (alerts)	✗	✓	✓
Citizen → incident → resource workflow	✓	✗	✗	✓	✓
AI-powered contextual disaster guidance (Online/Offline)	✗	✗	✗	✗	✓

✓ Fully available feature ✓ Partially available feature ✗ Feature not available

Key Research Insights

GPS-Based Disaster Reporting: Location-aware incident reporting.

Link - <https://www.sciencedirect.com/science/article/pii/S2590061719300523>

GIS-Based Disaster Management: Incident sharing & responder coordination.

Link - <https://www.sciencedirect.com/science/article/abs/pii/S0268401219311296>

Pocket RAG: Offline AI for emergency guidance.

Link - <https://arxiv.org/abs/2602.13229>

Existing Government Platforms

112 ERSS: Emergency reporting and response

SACHET: Geo-targeted disaster alerts & safety information

IDRN: Disaster-resource inventory and availability

Our System: Offline-first access, GPS SOS, offline AI guidance & incident queuing, with online synchronization for coordinated response.

